

ER DIAGRAM

1. Problem Statement

The Employee Burnout Prediction System is a machine-learning-based system designed to predict and monitor employee burnout risk. The system stores employee information, user details, machine learning model versions, prediction results, and the temporal features used for prediction.

The system allows authorized users to generate burnout predictions using a CNN-LSTM model. Each prediction is associated with an employee and a particular model version. Prediction features are stored to support the time-series analysis and future monitoring of employee burnout.

2. List of Identified Entities

The following entities are identified:

- ☐ Employee
- ☐ Employee_Burnout_Profile
- ☐ User
- ☐ Prediction
- ☐ Model
- ☐ Prediction_Feature

3. Entity Attributes

3.1 Employee

Stores basic employee and work-related information.

Attribute	Classification
employee_id	Primary Key
date_of_joining	Simple
gender	Simple
company_type	Simple
wfh_setup	Simple
designation	Simple
resource_allocation	Simple
mental_fatigue_score	Simple

PK: employee_id

3.2 Employee_Burnout_Profile

Stores the current burnout monitoring profile of an employee.

Attribute	Classification
profile_id	Primary Key
employee_id	Foreign Key
current_risk_level	Simple
last_assessment_date	Simple
monitoring_status	Simple

PK: profile_id

FK: employee_id

This entity represents the current burnout-monitoring profile of an employee.

3.3 User

Stores information about users who access the system.

Attribute	Classification
user_id	Primary Key
name	Simple
email	Simple
password	Simple
role	Simple

PK: user_id

The role can represent HR Manager, Administrator, or Machine Learning Engineer.

3.4 Prediction

Stores the burnout prediction generated by the system.

Attribute	Classification
prediction_id	Primary Key
employee_id	Foreign Key
model_id	Foreign Key
user_id	Foreign Key
prediction_date	Simple
risk_score	Simple
risk_level	Derived

PK: prediction_id

FKs: employee_id, model_id, user_id

Derived: risk_level

3.5 Model

Stores information about the machine-learning models used by the system.

Attribute	Classification
model_id	Primary Key
model_name	Simple
version	Simple
training_date	Simple
accuracy	Simple
status	Simple

PK: model_id

3.6 Prediction_Feature

Stores the feature values used for a particular prediction across different time periods.

Attribute	Classification
feature_id	Primary Key
prediction_id	Foreign Key
mental_fatigue	Simple
resource_allocation	Simple
designation	Simple
month	Simple

PK: feature_id

FK: prediction_id

4. Primary Keys and Foreign Keys

Entity	Primary Key	Foreign Keys
Employee	employee_id	—
Employee_Burnout_Profile	profile_id	employee_id
User	user_id	—
Prediction	prediction_id	employee_id, model_id, user_id
Model	model_id	—
Prediction_Feature	feature_id	prediction_id

5. Relationship Description

Relationship 1 — Employee and Employee_Burnout_Profile

Relationship: has_profile

Each employee has one current burnout profile, and each burnout profile belongs to exactly one employee.

Type: 1:1

Employee 1 ——— 1 Employee_Burnout_Profile

This provides the required One-to-One relationship.

Relationship 2 — Employee and Prediction

Relationship: has

One employee can have multiple predictions over time, while each prediction belongs to one employee.

Type: 1:M

Employee 1 ——— M Prediction

Relationship 3 — User and Prediction

Relationship: generates

One user can generate multiple predictions, while each prediction is generated by one user.

Type: 1:M

User 1 ——— M Prediction

Relationship 4 — Model and Prediction

Relationship: produces

One model version can produce multiple predictions, while each prediction uses one model version.

Type: 1:M

Model 1 ——— M Prediction

Relationship 5 — Prediction and Prediction_Feature

Relationship: contains

One prediction contains multiple feature records, while each feature record belongs to one prediction.

Type: 1:M

Prediction 1 ——— M Prediction_Feature

6. Many-to-Many Relationship

This is the important part for your requirement.

Employee ↔ Model

An employee can be evaluated using multiple versions of the machine-learning model.

At the same time, one model version can be used to generate predictions for many employees.

Therefore:

Employee M ——— N Model

This is an M:N relationship.

However, we should not directly draw an M:N line between Employee and Model.

Instead, the relationship is resolved using the associative entity:

Prediction

So:

Employee

1

|

M

Prediction

M

|

1

Model

Therefore, Prediction acts as the bridge/associative entity between Employee and Model.

Why Prediction is a good bridge entity

The relationship itself has important attributes:

- ☐ prediction_id
- ☐ prediction_date
- ☐ risk_score
- ☐ risk_level
- ☐ user_id

Therefore, Prediction is not just a technical bridge; it represents an actual prediction event in your system.

7. Cardinality and Participation Constraints

Relationship	Type	Cardinality	Participation
Employee – Employee_Burnout_Profile	1:1	1:1	Employee: Partial, Profile: Total
Employee – Prediction	1:M	1:M	Employee: Partial, Prediction: Total
User – Prediction	1:M	1:M	User: Partial, Prediction: Total
Model – Prediction	1:M	1:M	Model: Partial, Prediction: Total
Prediction – Prediction_Feature	1:M	1:M	Prediction: Total, Feature: Total
Employee – Model	M:N	M:N	Resolved through Prediction

Explanation

- ☐ An employee may or may not have a burnout profile, but each profile belongs to one employee.
- ☐ An employee may have multiple predictions.
- ☐ Every prediction must belong to one employee.
- ☐ A user can generate multiple predictions.
- ☐ Every prediction is generated by one user.
- ☐ A model can generate multiple predictions.
- ☐ Every prediction uses one model.
- ☐ A prediction contains multiple temporal feature records.
- ☐ An employee can be associated with multiple model versions through predictions.
- ☐ A model can be associated with multiple employees through predictions.

8. Attribute Classification

Simple Attributes

Examples:

- ☐ gender
- ☐ company_type
- ☐ designation
- ☐ resource_allocation
- ☐ mental_fatigue_score
- ☐ model_name
- ☐ version
- ☐ role

Composite Attributes

- ☐ current_risk_level
- ☐ name
- ☐ designation

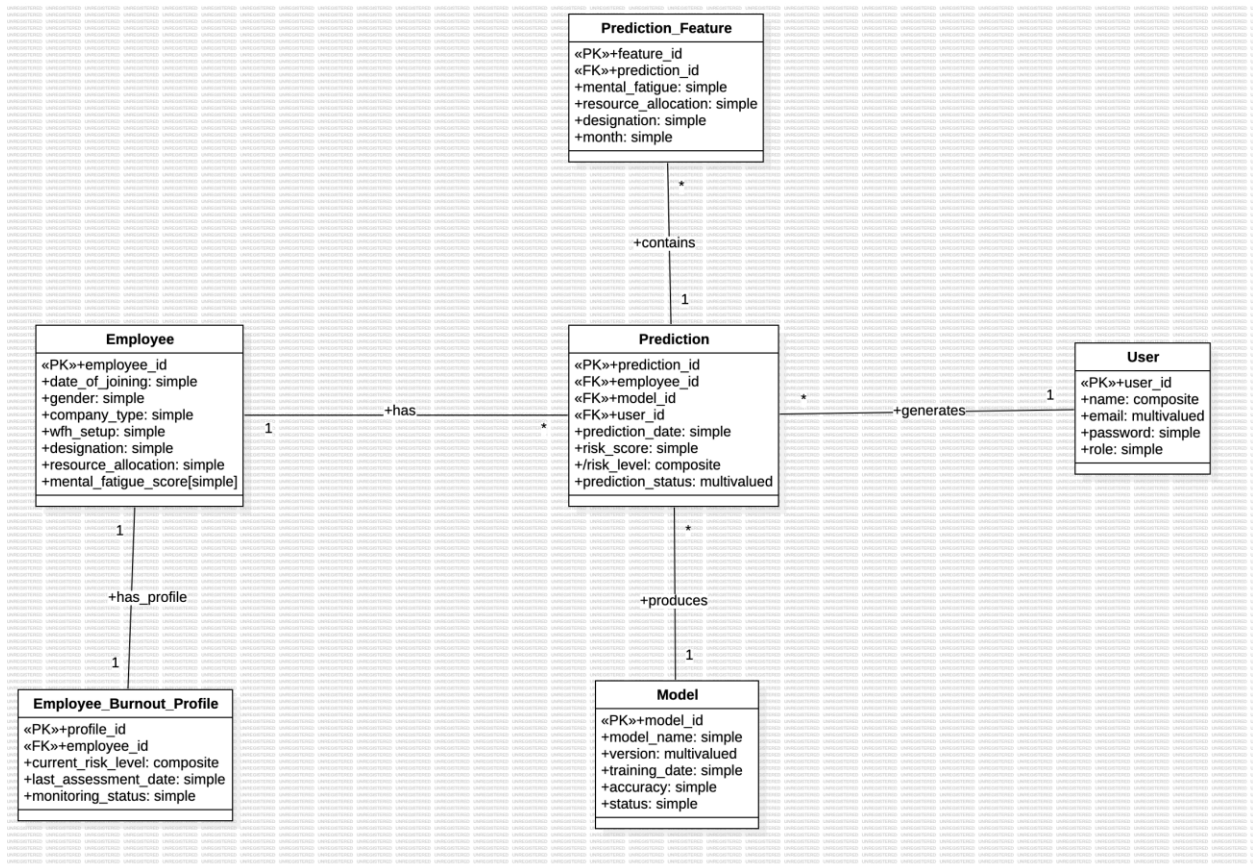
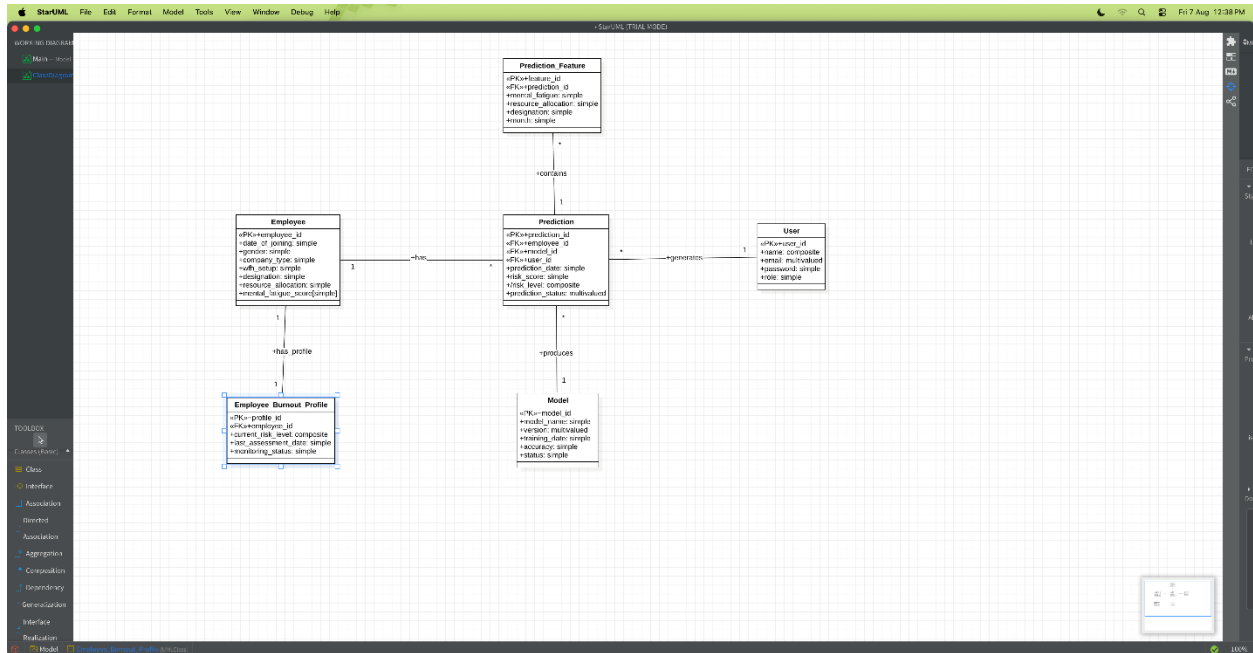
Multivalued Attributes

- ☐ prediction_status
- ☐ version
- ☐ designation
- ☐ email

Derived Attribute

risk_level is derived from risk_score.

9. Final ER Diagram Structure



10. Brief Explanation of the Design

The ER diagram represents the database structure of the Employee Burnout Prediction System. The Employee entity stores employee information, while Employee_Burnout_Profile maintains the employee's current burnout monitoring information through a 1:1 relationship. Prediction records connect employees, users, and machine-learning models and therefore resolve the M:N relationship between Employee and Model. Prediction_Feature stores the temporal feature values used for CNN-LSTM predictions. Primary and foreign keys maintain entity integrity and establish relationships between the entities.